Exercise J-M

1. The set of all values of λ for which the equation $\cos^2 2x - 2\sin^4 x - 2\cos^2 x = \lambda$ has a real solution x, is :

[JEE(Main)Jan.2023]

λ के सभी मानों, जिनके लिए समीकरण $\cos^2 2x - 2\sin^4 x - 2\cos^2 x = \lambda$ का एक वास्तविक हल x है, का समुच्चय है :

[JEE(Main)Jan.2023]

(A) $\left[-1, -\frac{1}{2}\right]$ (B) $\left[-2, -\frac{3}{2}\right]$ (C) [-2,-1] (D) $\left[-\frac{3}{2}, -1\right]$

Ans. D

Sol. $\lambda = \cos^2 2x - 2\sin^4 x - 2\cos^2 x$

convert all in to cos x.

$\lambda = (2\cos^2 x - 1)^2 - 2(1 - \cos^2 x)^2 - 2\cos^2 x$

$= 4\cos^4 x - 4\cos^2 x + 1 - 2(1 - 2\cos^2 x + \cos^4 x) - 2\cos^2 x$

$= 2\cos^4 x - 2\cos^2 x + 1 - 2$

$= 2\cos^4 x - 2\cos^2 x - 1$

$$= 2\left[\cos^4 x - \cos^2 x - \frac{1}{2}\right]$$

$$= 2\left[\left(\cos^2 x - \frac{1}{2}\right)^2 - \frac{3}{4}\right]$$

$$\lambda_{max} = 2\left[\frac{1}{4} - \frac{3}{4}\right] = 2 \times \left(-\frac{2}{4}\right) = -1 \text{ (max Value)}$$

$$\lambda_{min} = 2\left[0 - \frac{3}{4}\right] = -\frac{3}{2} \text{ (minimum value)}$$

$$\text{So, Range} = \left[-\frac{3}{2}, -1\right]$$

2. If the solution of the equation $\log_{\cos x}\cot x + 4\log_{\sin x}\tan x = 1$, $x \in \left(0, \frac{\pi}{2}\right)$, is $\sin^{-1}\left(\frac{\alpha + \sqrt{\beta}}{2}\right)$ where α, β are integers, then $\alpha + \beta$ is equal to :

[JEE(Main)Jan.2023]

यदि समीकरण $\log_{\cos x}\cot x + 4\log_{\sin x}\tan x = 1$, $x \in \left(0, \frac{\pi}{2}\right)$ का हल $\sin^{-1}\left(\frac{\alpha + \sqrt{\beta}}{2}\right)$ हैं, जहाँ α, β पूर्णांक है, तो $\alpha + \beta$ बराबर है :

[JEE(Main)Jan.2023]

(A) 6 (B) 3 (C) 4 (D) 5

Ans. C

Sol. $\log_{\cos x}\cot x + 4\cdot\log_{\sin x}\tan x = 1$

$$\frac{\log\cos x - \log\sin x}{\log\cos x} + 4\frac{(\log\sin x - \log\cos x)}{\log\sin x} = 1$$

$(\log \sin x)^2 - 4(\log \sin x)(\log \cos x) + 4(\log \cos x)^2 = 1$

$\log \sin x = 2 \log \cos x$

$\sin^2 x + \sin x - 1 = 0$

$$\sin x = \frac{-1 \pm \sqrt{5}}{2}$$

$\alpha = -1, \beta = 5$

$\alpha + \beta = 4$

3. Let $S = \{\theta \in [0, 2\pi) : \tan(\pi \cos\theta) + \tan(\pi \sin\theta) = 0\}$. Then $\sum_{\theta\in S} \sin^2\left(\theta + \frac{\pi}{4}\right)$ is equal to _____.

[JEE(Main)Jan.2023]

माना $S = \{\theta \in [0, 2\pi) : \tan(\pi \cos\theta) + \tan(\pi \sin\theta) = 0\}$ है। तब $\sum_{\theta\in S} \sin^2\left(\theta + \frac{\pi}{4}\right)$ बराबर है _____

[JEE(Main)Jan.2023]

Ans. 2

Sol. $\tan(\pi\cos\theta) + \tan(\pi\sin\theta) = 0$

$\tan(\pi\cos\theta) = -\tan(\pi\sin\theta)$

$\tan(\pi\cos\theta) = \tan(-\pi\sin\theta)$

$\pi\cos\theta = n\pi - \pi\sin\theta, n \in I$

$\sin\theta + \cos\theta = n$

$\therefore \sin\theta + \cos\theta \in \left[-\sqrt{2}, \sqrt{2}\right]$

Then possible value of $n = -1, 0, 1$

Now $\sin\theta + \cos\theta = 1$

$$\theta = 0, \frac{\pi}{2},$$

$$\left|\begin{array}{c} \sin\theta + \cos\theta = 0 \\ \tan\theta - 1 \\ \theta = \frac{3\pi}{4}, \frac{7\pi}{4}, \end{array}\right|$$

$\sin\theta + \cos\theta = -1$

$\theta = \frac{3\pi}{2}, \pi$

So, $\sum_{\theta \in S} \sin^2\left(\theta + \frac{\pi}{4}\right) = 4\left(\frac{1}{2}\right) + 2(0) = 2$

4. If m and n respectively are the numbers of positive and negative values of θ in the interval $[-\pi, \pi]$ that satifsy the equation $\cos 2\theta \cos\frac{\theta}{2} = \cos 3\theta \cos\frac{9\theta}{2}$, then mn is equal to _____. **[JEE(Main)Jan.2023]**

यदि समीकरण $\cos 2\theta \cos\frac{\theta}{2} = \cos 3\theta \cos\frac{9\theta}{2}$ को संतुष्ट करने वाले अंतराल $[-\pi, \pi]$ में θ के धनात्मक तथा ऋणात्मक मानों की संख्या क्रमशः m तथा n है, तो mn बराबर है। **[JEE(Main)Jan.2023]**

Ans. 25

Sol. $\cos\frac{5\theta}{2} + \cos\frac{3\theta}{2} = \cos\frac{15\theta}{2} + \cos\frac{3\theta}{1} \Rightarrow \frac{15\theta}{2} = 2n\pi \pm \frac{5\theta}{2}$ $\quad 5\theta = 2n\pi \Rightarrow \frac{2n\pi}{5}$ $\quad \theta = \frac{n\pi}{5}$

$10\theta = 2n\pi \Rightarrow \frac{n\pi}{5}$

$-\pi \le \frac{p\pi}{5} \le \pi$

$-5 \le p \le 5$

$m = 5$

$n = 5$

$mn = 25$

5. Let $S = \left\{x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right) : 9^{1-\tan^2 x} + 9^{\tan^2 x} = 10\right\}$ and $\beta = \sum_{x \in S} \tan^2\left(\frac{x}{3}\right)$, then $\frac{1}{6}(\beta - 14)^2$ is equal to :

[JEE(Main)April,2023]

माना $S = \left\{x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right) : 9^{1-\tan^2 x} + 9^{\tan^2 x} = 10\right\}$ तथा $\beta = \sum_{x \in S} \tan^2\left(\frac{x}{3}\right)$ हैं, तो $\frac{1}{6}(\beta - 14)^2$ बराबर है :

[JEE(Main)April,2023]

(A) 32 (B) 64 (C) 16 (D) 8

Ans. A

Sol. $9^{1-\tan^2 x} + 9^{\tan^2 x} = 10$

$\Rightarrow \frac{9}{y} + y = 10 \ \left(\text{where } 9^{\tan^2 x} = y\right)$

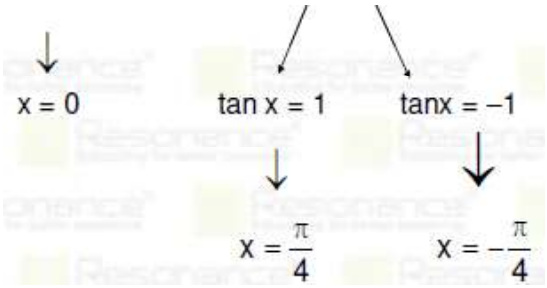
So, $9 + y^2 = 10y$

$y^2 - 10y + 9 = 0$

$(y-1)(y-9) = 0$

So, $9^{\tan^2 x} = 1$ & $9^{\tan^2 x} = 9$

$\tan^2 x = 0$ $\tan^2 x = 1$



Now $\beta = \sum \tan^2 \frac{x}{3} = \tan^2 \frac{0}{3} + \tan^2 \frac{\pi}{12} + \tan^2 \frac{\pi}{12}$

$= 2\tan^2 \frac{\pi}{12}$

$= 2(2-\sqrt{3})^2$

$= 2(7-4\sqrt{3})$

$\beta = 14 - 8\sqrt{3} \Rightarrow \frac{1}{6}(\beta - 14)^2 = 32$

6. The number of elements in the set $S = \{\theta \in [0, 2\pi] : 3\cos^4\theta - 5\cos^2\theta - 2\sin^6\theta + 2 = 0\}$ is :

[JEE(Main)April,2023]

समुच्चय $S = \{\theta \in [0, 2\pi] : 3\cos^4\theta - 5\cos^2\theta - 2\sin^6\theta + 2 = 0\}$ में अवयवों की संख्या है :

[JEE(Main)April,2023]

(A) 9 (B) 10 (C) 8 (D) 12

Ans. A

Sol. $3\cos^4\theta - 5\cos^2\theta - 2\sin^6\theta + 2 = 0$

$\Rightarrow 3\cos^4\theta - 3\cos^2\theta - 2\cos^2\theta - 2\sin^6\theta + 2 = 0$

$\Rightarrow 3\cos^4\theta - 3\cos^2\theta + 2\sin^2\theta - 2\sin^6\theta = 0$

$\Rightarrow 3\cos^2\theta\left(\cos^2\theta - 1\right) + 2\sin^2\theta\left(\sin^4\theta - 1\right) = 0$

$\Rightarrow -3\cos^2\theta\sin^2\theta + 2\sin^2\theta\left(1 + \sin^2\theta\right)\cos^2\theta - 1$

$\Rightarrow \sin^2\theta\cos^2\theta\left(2 + 2\sin^2\theta - 3\right) = 0$

$\Rightarrow \sin^2\theta\cos^2\theta\left(2\sin^2\theta - 1\right) = 0$

(C1) $\sin^2\theta = 0 \rightarrow 3$ solution; $\theta = \{0, \pi, 2\pi\}$

(C2) $\cos^2\theta = 0 \rightarrow 2$ solution; $\theta = \left\{\frac{\pi}{2}, \frac{3\pi}{2}\right\}$

(C3) $\sin^2\theta = \frac{1}{2} \rightarrow 4$ solution; $\theta = \left\{\frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{7\pi}{4}\right\}$

No. of solution = 9

7. If $2\tan^2\theta - 5\sec\theta = 1$ has exactly 7 solutions in the interval $\left[0, \frac{n\pi}{2}\right]$, for the least value of $n \in N$, then $\sum_{k=1}^{n} \frac{k}{2^k}$ is equal to : **[JEE(Main)Jan.2024]**

जिस $n \in N$ के न्यूनतम मान के लिए अंतराल $\left[0, \frac{n\pi}{2}\right]$ में $2\tan^2\theta - 5\sec\theta = 1$ के ठीक 7 हल हैं, उस n के लिए $\sum_{k=1}^{n} \frac{k}{2^k}$ बराबर है : **[JEE(Main)Jan.2024]**

(A) $\frac{1}{2^{14}}\left(2^{15} - 15\right)$ (B) $1 - \frac{15}{2^{13}}$ (C) $\frac{1}{2^{13}}\left(2^{14} - 15\right)$ (D) $\frac{1}{2^{15}}\left(2^{14} - 14\right)$

Ans. C

Sol. $2\tan^2\theta - 5\sec\theta - 1 = 0$

$\Rightarrow 2\sec^2\theta - 5\sec\theta - 3 = 0$

$\Rightarrow (2\sec\theta + 1)(\sec\theta - 3) = 0$

$\Rightarrow \sec\theta = -\frac{1}{2}, 3$

$\Rightarrow \cos\theta = -2, \frac{1}{3}$

$\Rightarrow \cos\theta = \frac{1}{3}$

For 7 solutions n = 13

So, $\sum_{k=1}^{13} \frac{k}{2^k} = S$ (say)

$S = \frac{1}{2} + \frac{2}{2^2} + \frac{3}{2^3} + \ldots + \frac{13}{2^{13}}$

$\frac{1}{2}S = \frac{1}{2^2} + \frac{1}{2^3} + \ldots + \frac{12}{2^{13}} + \frac{13}{2^{14}}$

$\Rightarrow \frac{S}{2} = \frac{1}{2} \cdot \frac{1 - \frac{1}{2^{13}}}{1 - \frac{1}{2}} - \frac{13}{2^{14}}$

$$\Rightarrow S = 2\cdot\left(\frac{2^{13}-1}{2^{13}}\right)-\frac{13}{2^{13}}$$

8. If $\alpha, -\frac{\pi}{2} < \alpha < \frac{\pi}{2}$ is the solution of $4\cos\theta + 5\sin\theta = 1$, then the value of $\tan\alpha$ is :

[JEE(Main)Jan.2024]

यदि समीकरण $4\cos\theta + 5\sin\theta = 1$ का हल $\alpha, -\frac{\pi}{2} < \alpha < \frac{\pi}{2}$ है, तो $\tan\alpha$ का मान है : **[JEE(Main)Jan.2024]**

(A) $\frac{10-\sqrt{10}}{12}$ (B) $\frac{\sqrt{10}-10}{6}$ (C) $\frac{10-\sqrt{10}}{6}$ (D) $\frac{\sqrt{10}-10}{12}$

Ans. D

Sol. $4 + 5\tan\theta = \sec\theta$

Squaring : $24\tan^2\theta + 40\tan\theta + 15 = 0$

$$\tan\theta = \frac{-10\pm\sqrt{10}}{12}$$

and $\tan\theta = -\left(\frac{10+\sqrt{10}}{12}\right)$ is Rejected

9. The sum of the solutions $x \in R$ of the equation $\frac{3\cos 2x + \cos^3 2x}{\cos^6 x - \sin^6 x} = x^3 - x^2 + 6$ is :

[JEE(Main)Jan.2024]

समीकरण $\frac{3\cos 2x + \cos^3 2x}{\cos^6 x - \sin^6 x} = x^3 - x^2 + 6$ के हलों $x \in R$ का योगफल है : **[JEE(Main)Jan.2024]**

(A) 1 (B) 0 (C) –1 (D) 3

Ans. A

Sol. Given $\frac{3\cos 2x + \cos^3 2x}{\cos^6 x - \sin^6 x} = x^3 - x^2 + 6$

$$\Rightarrow \frac{\cos 2x\left(3+\cos^2 2x\right)}{\left(\cos^2 x - \sin^2 x\right)\left[\sin^4 x + \cos^4 x + \sin^2 x\cos^2 x\right]}$$

$$= x^3 - x^2 + 6$$

$$\Rightarrow \frac{3+\cos^2 2x}{1-\sin^2 x\cos^2 x} = x^3 - x^2 + 6$$

$$\Rightarrow 4\left(\frac{3+\cos^2 2x}{4-\sin^2 2x}\right) = x^3 - x^2 + 6$$

$\Rightarrow x^3 - x^2 + 2 = 0$

$\Rightarrow (x+1)\left(x^2 - 2x + 2\right) = 0$

$\Rightarrow (x+1)\left((x-1)^2 + 1\right) = 0$

$\therefore$ Sum of solutions $= -1$

10. If $2\sin^3 x + \sin 2x \cos x + 4 \sin x - 4 = 0$ has exactly 3 solutions in the interval $\left[0, \frac{n\pi}{2}\right]$, $n \in N$, then the roots of the equation $x^2 + nx + (n-3) = 0$ belongs to : **[JEE(Main)Jan.2024]**

यदि $2\sin^3 x + \sin 2x \cos x + 4 \sin x - 4 = 0$ के अंतराल $\left[0, \frac{n\pi}{2}\right]$, $n \in N$ में ठीक 3 हल हैं, तो समीकरण $x^2 + nx + (n-3) = 0$ के मूल किस में हैं? **[JEE(Main)Jan.2024]**

(A) $(-\infty, 0)$ (B) $\left(-\frac{\sqrt{17}}{2}, \frac{\sqrt{17}}{2}\right)$ (C) $(0, \infty)$ (D) Z

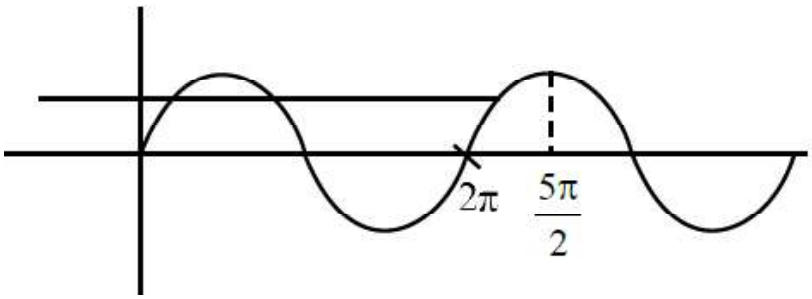
Ans. A

Sol. $2\sin^3 x + 2 \sin x \cos^2 x + 4 \sin x - 4 = 0$

$\Rightarrow 2\sin^3 x + 2\sin x (1 - \sin^2 x) + 4 \sin x - 4 = 0$

$\Rightarrow 2\sin^3 x + 6 \sin x - 2 \sin^3 x - 4 = 0$

$\sin x = \frac{2}{3}$



$\left[0, \frac{5\pi}{2}\right]$

$x^2 + 5x + 2 = 0$

$\alpha, \beta = \frac{-5 \pm \sqrt{17}}{2}$ both negative, hence $(-\infty, 0)$

11. The number of solutions, of the equation $e^{\sin x} - 2e^{-\sin x} = 2$, is : **[JEE(Main)Jan.2024]**

(A) 0 (B) more than 2 (C) 2 (D) 1

समीकरण $e^{\sin x} - 2e^{-\sin x} = 2$ के हलों की संख्या है : **[JEE(Main)Jan.2024]**

(A) 0 (B) more than 2 (C) 2 (D) 1

Ans. A

Sol. $e^{\sin x} - 2e^{-\sin x} = 2$

Let $y = e^{\sin x}$, $e^{\sin x} > 0 \forall x \Rightarrow y > 0$

$\Rightarrow y - \frac{2}{y} = 2 \Rightarrow y^2 - 2y - 2 = 0$

$\Rightarrow y = \sqrt{3} + 1$

$\Rightarrow e^{\sin x} = \sqrt{3} + 1$

$\sin x = \ell n\left(\sqrt{3} + 1\right) > 1$

$\Rightarrow$ No solution $\Rightarrow$ Number of solution is zero

12. The number of solutions of the equation $4\sin^2 x - 4\cos^3 x + 9 - 4\cos x = 0; x \in [-2\pi, 2\pi]$ is : **[JEE(Main)Jan.2024]**

समीकरण $4\sin^2 x - 4\cos^3 x + 9 - 4\cos x = 0; x \in [-2\pi, 2\pi]$ के हलों की संख्या है : **[JEE(Main)Jan.2024]**

(A) 0 (B) 3 (C) 1 (D) 2

Ans. A

Sol. $4 - 4\cos^2 x - 4\cos^3 x + 9 - 4\cos x = 0$

$4\cos^3 x + 4\cos^2 x + 4\cos x = 13$

$\because$ LHS ≤ 12

So equation will have no solution.

13. Let the set of all $a \in R$ such that the equation $\cos 2x + a \sin x = 2a - 7$ has a solution be [p, q] and $r = \tan 9^\circ - \tan 27^\circ - \frac{1}{\cot 63^\circ} + \tan 81^\circ$, then pqr is equal to _____. **[JEE(Main)Jan.2024]**

माना $a \in R$ के सभी मानों, जिनके लिए समीकरण $\cos 2x + a \sin x = 2a - 7$ का एक हल है, का समुच्चय [p, q] है तथा $r = \tan 9^\circ - \tan 27^\circ - \frac{1}{\cot 63^\circ} + \tan 81^\circ$ है। तो pqr बराबर है _____। **[JEE(Main)Jan.2024]**

Ans. 48

Sol. $\cos 2x + a \cdot \sin x = 2a - 7$

$a(\sin x - 2) = 2(\sin x - 2)(\sin x + 2)$

$\sin x = 2, a = 2(\sin x + 2)$

$\Rightarrow a \in [2, 6]$

$p = 2 \quad q = 6$

$r = \tan 9^\circ + \cot 9^\circ - \tan 27 - \cot 27$

$r = \frac{1}{\sin 9 \cdot \cos 9} - \frac{1}{\sin 27 \cdot \cos 27}$

$= 2\left[\frac{4}{\sqrt{5}-1} - \frac{4}{\sqrt{5}+1}\right]$

$r = 4$

$p \cdot q \cdot r = 2 \times 6 \times 4 = 48$

14. Suppose $\theta \in \left[0, \frac{\pi}{4}\right]$ is a solutions of $4\cos\theta - 3\sin\theta = 1$. Then $\cos\theta$ is equal to :

[JEE(Main)April,2024]

माना $4\cos\theta - 3\sin\theta = 1$ का एक हल $\theta \in \left[0, \frac{\pi}{4}\right]$ है। तो $\cos\theta$ बराबर है : **[JEE(Main)April,2024]**

(A) $\frac{4}{\left(3\sqrt{6}+2\right)}$ (B) $\frac{4}{\left(3\sqrt{6}-2\right)}$ (C) $\frac{6-\sqrt{6}}{\left(3\sqrt{6}-2\right)}$ (D) $\frac{6+\sqrt{6}}{\left(3\sqrt{6}+2\right)}$

Ans. B

Sol. $4\cos\theta - 3\sin\theta = 1$

$4\cos\theta - 1 = 3\sin\theta$

$16\cos^2\theta + 1 - 8\cos\theta = 9(1 - \cos^2\theta)$

$\Rightarrow 25\cos^2\theta - 8\cos\theta - 8 = 0$

$\Rightarrow \cos\theta = \frac{8 \pm \sqrt{64 + 4 \times 25 \times 8}}{2.25}$

$= \frac{8 \pm 4\sqrt{4 + 50}}{2.25}$

$= \frac{4 \pm 2\sqrt{54}}{25}$

As $\theta \in \left[0, \frac{\pi}{4}\right]$

$\Rightarrow \cos\theta = \frac{4 + 6\sqrt{6}}{25} = \frac{4}{3\sqrt{6} - 2}$

15. Let $S = \{\sin^2 2\theta : (\sin^4\theta + \cos^4\theta)x^2 + (\sin 2\theta)x + (\sin^6\theta + \cos^6\theta) = 0$ has real roots$\}$. If α and β be the smallest and largest elements of the set S, respectively, then $3((\alpha - 2)^2 + (\beta - 1)^2)$ equals _____.

[JEE(Main)April,2024]

माना $S = \{\sin^2 2\theta : (\sin^4\theta + \cos^4\theta)x^2 + (\sin 2\theta)x + (\sin^6\theta + \cos^6\theta) = 0$ के वास्तविक मूल हैं$\}$ हैं। यदि α तथा β समुच्चय S

के क्रमशः सबसे छोटा तथा सबसे बड़ा अवयव हैं, तो $3((\alpha-2)^2+(\beta-1)^2)$ बराबर है _____

[JEE(Main)April,2024]

Ans. 4

Sol. $D \geq 0$

$$\Rightarrow \sin^2 2\theta - 4\left(\sin^4\theta+\cos^4\theta\right)\left(\sin^6\theta+\cos^6\theta\right) \geq 0$$

$$\Rightarrow t - 4\left(1-\frac{t}{2}\right)\left(1-\frac{3t}{4}\right) \geq 0 \qquad (\text{Let } \sin^2 2\theta = t)$$

$$\Rightarrow 3t^2 - 12t + 8 \leq 0$$

$$\Rightarrow t \in \left[2-\frac{2}{\sqrt{3}}, 2+\frac{2}{\sqrt{3}}\right] \qquad \text{_____(1)}$$

But $t = \sin^2 2\theta \in [0, 1]$ _____(2)

$$(A) \cap (2) t \in \left[2-\frac{2}{\sqrt{3}}, 1\right]$$

$$\alpha = 2-\frac{2}{\sqrt{3}},\ \beta = 1$$

$$3((\alpha-2)^2+(\beta-1)^2) = 4$$

16. The number of solutions of $\sin^2 x + (2 + 2x - x^2)\sin x - 3(x-1)^2 = 0$, where $-\pi \leq x \leq \pi$, is _____.

[JEE(Main)April,2024]

$\sin^2 x + (2 + 2x - x^2)\sin x - 3(x-1)^2 = 0$, जहाँ $-\pi \leq x \leq \pi$ के हलों की संख्या है _____

[JEE(Main)April,2024]r

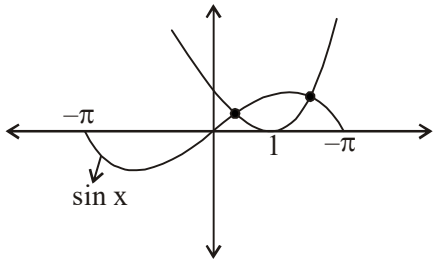
Ans. 2

Sol. $\sin^2 x + (3 - (x-1)^2)\sin x - 3(x-1)^2 = 0$

$\sin^2 x + 3\sin x - (x-1)^2 \sin x - 3(x-1)^2 = 0$

$\sin x(\sin x + 3) - (x-1)^2[\sin x + 3] = 0$

$$\underbrace{(\sin x + 3)}_{\substack{\sin x \neq -3 \\ \downarrow \\ \text{not possible}}} \qquad \underbrace{\left(\sin x - (x-1)^2\right)}_{\sin x = (x-1)^2} = 0$$



There are two intersections between this graph.

So, number of solution will be 2.